

DNS

- Locate the DNS query and response messages. Are they sent over UDP or TCP?

The image shows a Wireshark packet capture window. The top pane displays a list of packets. Packet 307 is highlighted, showing a DNS Standard query from 172.16.4.180 to 172.16.1.4. The bottom pane shows the details of packet 307, including the User Datagram Protocol (UDP) and Domain Name System (DNS) sections. The DNS section shows a query for wpad.calicut.nielit.in.

No.	Time	Source	Destination	Protocol	Length	Info
297	4.710910	172.16.1.32	239.255.255.250	SSDP	412	NOTIFY * HTTP/1.1
298	4.711777	172.16.1.31	239.255.255.250	SSDP	412	NOTIFY * HTTP/1.1
299	4.711777	172.16.1.33	239.255.255.250	SSDP	420	NOTIFY * HTTP/1.1
300	4.711777	172.16.1.32	239.255.255.250	SSDP	420	NOTIFY * HTTP/1.1
301	4.712046	172.16.1.34	239.255.255.250	SSDP	420	NOTIFY * HTTP/1.1
302	4.712576	172.16.1.31	239.255.255.250	SSDP	420	NOTIFY * HTTP/1.1
303	5.242042	172.16.4.180	104.16.45.99	TLSv1.2	201	Application Data
304	5.242222	172.16.4.180	104.16.45.99	TLSv1.2	93	Application Data
305	5.254859	104.16.45.99	172.16.4.180	TCP	54	443 → 62228 [ACK] Seq=1 Ack=148 Win=513 Len=0
306	5.254859	104.16.45.99	172.16.4.180	TCP	54	443 → 62228 [ACK] Seq=1 Ack=187 Win=513 Len=0
307	5.255313	172.16.4.180	172.16.1.4	DNS	82	Standard query 0x29e4 A wpad.calicut.nielit.in
308	5.261781	172.16.1.4	172.16.4.180	DNS	127	Standard query response 0x29e4 No such name A wpad.calicut.nielit.in SOA ns.calicut.nielit.in
309	5.364834	172.16.4.191	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
310	5.364834	104.16.45.99	172.16.4.180	TLSv1.2	93	Application Data
311	5.364834	104.16.45.99	172.16.4.180	TLSv1.2	328	Application Data
312	5.365137	104.16.45.99	172.16.4.180	TCP	1462	443 → 62228 [PSH, ACK] Seq=314 Ack=187 Win=514 Len=1408 [TCP segment of a reassembled PDU]
313	5.365137	104.16.45.99	172.16.4.180	TCP	1462	443 → 62228 [PSH, ACK] Seq=1722 Ack=187 Win=514 Len=1408 [TCP segment of a reassembled PDU]

Frame 307: 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Device\NPF_{6C29EF...}

Ethernet II, Src: CloudNet_3f:c3:0b (30:03:c8:3f:c3:0b), Dst: Broadcom_1e:5f:a2 (00:0a:f7:1e:5f:a2)

Internet Protocol Version 4, Src: 172.16.4.180, Dst: 172.16.1.4

User Datagram Protocol, Src Port: 54434, Dst Port: 53

Source Port: 54434

Destination Port: 53

Length: 48

Checksum: 0x741c [unverified]

[Checksum Status: Unverified]

[Stream index: 10]

[Timestamps]

UDP payload (40 bytes)

Domain Name System (query)

Transaction ID: 0x29e4

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

wpad.calicut.nielit.in: type A, class IN

[Response In: 308]

- What is the destination port for the DNS query message? What is the source port of the DNS response message?

User Datagram Protocol, Src Port: 54434, Dst Port: 53

Source Port: 54434

Destination Port: 53

Length: 48

- To what IP address is the DNS query message sent?

The image shows a Wireshark packet capture window and a Command Prompt window. The Wireshark packet capture shows a list of packets. Packet 307 is highlighted, showing a DNS Standard query from 172.16.4.180 to 172.16.1.4. The Command Prompt window shows the output of the 'ipconfig /all' command, displaying the DNS Servers as 172.16.1.4 and NetBIOS over Tcpip as Enabled.

No.	Time	Source	Destination	Protocol	Length	Info
304	5.242222	172.16.4.180	104.16.45.99	TLSv1.2	93	Application Data
305	5.254859	104.16.45.99	172.16.4.180	TCP	54	443 → 62228 [ACK] Seq=1 Ack=148 Win=513 Len=0
306	5.254859	104.16.45.99	172.16.4.180	TCP	54	443 → 62228 [ACK] Seq=1 Ack=187 Win=513 Len=0
307	5.255313	172.16.4.180	172.16.1.4	DNS	82	Standard query 0x29e4 A wpad.calicut.nielit.in
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309	5.364834	172.16.4.191	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
310	5.364834	104.16.45.99	172.16.4.180	TLSv1.2	93	Application Data

Command Prompt

DNS Servers : 172.16.1.4

NetBIOS over Tcpip. : Enabled

C:\Users\dell>

- Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?

The image shows the details of a DNS query message in Wireshark. The Domain Name System (query) section is expanded, showing the Transaction ID, Flags, Questions, Answer RRs, Authority RRs, Additional RRs, and Queries. The query is for wpad.calicut.nielit.in: type A, class IN.

Domain Name System (query)

Transaction ID: 0x29e4

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

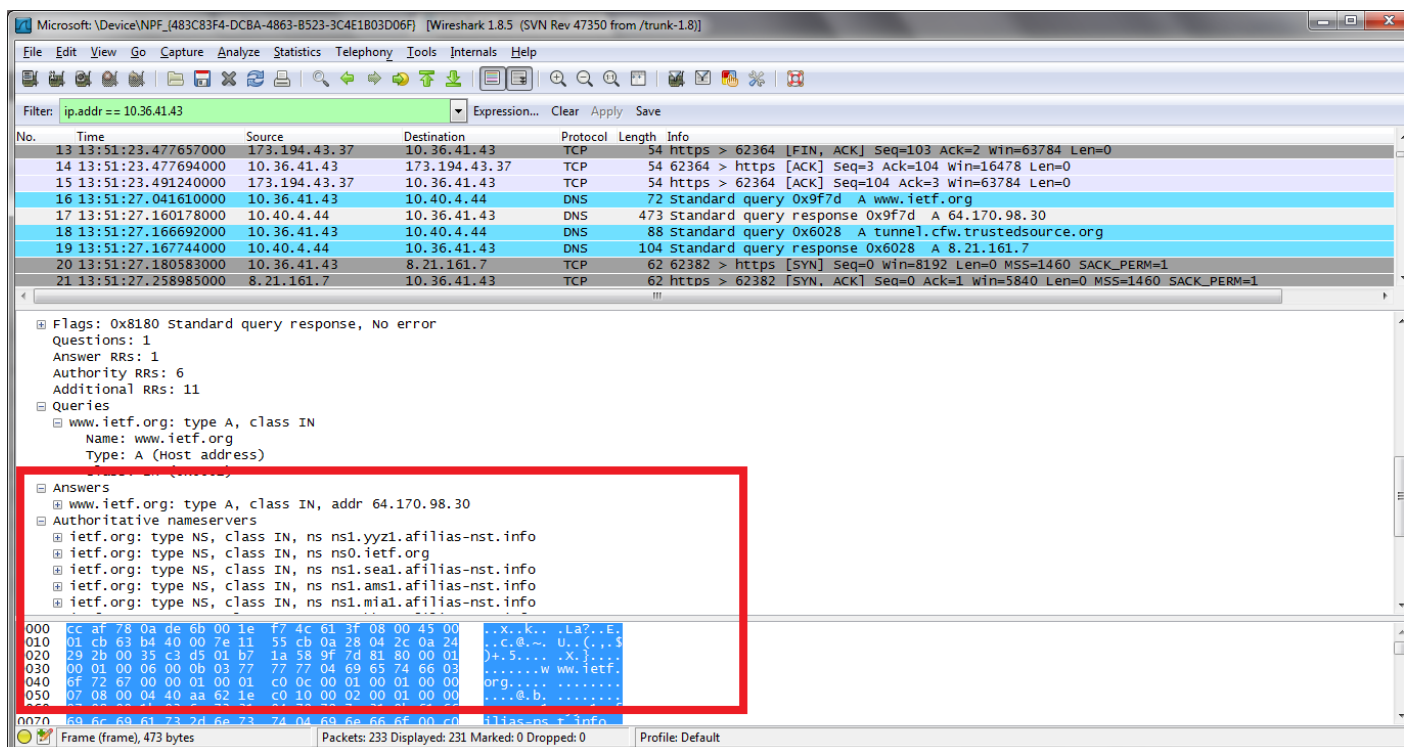
Additional RRs: 0

Queries

wpad.calicut.nielit.in: type A, class IN

[Response In: 308]

- Examine the DNS response message. How many “Answers, Authoritative & Additional” are provided? What does each of these answers contain? Provide a screenshot of all the above questions with answers



TCP/IP

- Which field indicates the version of Internet protocol in use?

Internet Protocol Version 4, Src: 172.16.4.180, Dst: 172.16.1.4
0100 = Version: 4

- What are the Source and destination IP addresses of the IP Packet?

Src: 172.16.4.180, Dst: 172.16.1.4

- What is the IP Header Length and what is the size of the field used to indicate it?

```

v Internet Protocol Version 4, Src: 172.16.4.180, Dst: 172.16.1.4
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 68
    Identification: 0xab47 (43847)

```

- What is the Length of the IP packet whose source address is your IP? (Choose any one packet)

```

  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 68
    Identification: 0xab47 (43847)

```

- Which field indicates whether the IP Packet has been fragmented or not and how?

the "Fragmentation Flags" field in the IP header, along with the "Fragment Offset" field, indicates whether the IP packet has been fragmented and provides the necessary information for proper reassembly when fragments arrive at their destination.

```

  > 000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0

```

- What is the TTL value of the IP packet?

```

  ...0 0000 0000 0000 =
    Time to Live: 128
    Protocol: UDP (17)

```

- Which field verifies the integrity of the IP Header?

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    Header Checksum: 0x3189 [validation disabled]
    [Header checksum status: Unverified]

```